

Example Process for Designing a Hybrid User Interface for a Multi-Robot System

Jan P. Gründling*
University of Trier

Nathalie Schauffel
University of Trier

Simon Oehrl
RWTH Aachen

Sebastian Pape
RWTH Aachen

Torsten W. Kühlen
RWTH Aachen

Thomas Ellwart
University of Trier

Benjamin Weyers
University of Trier

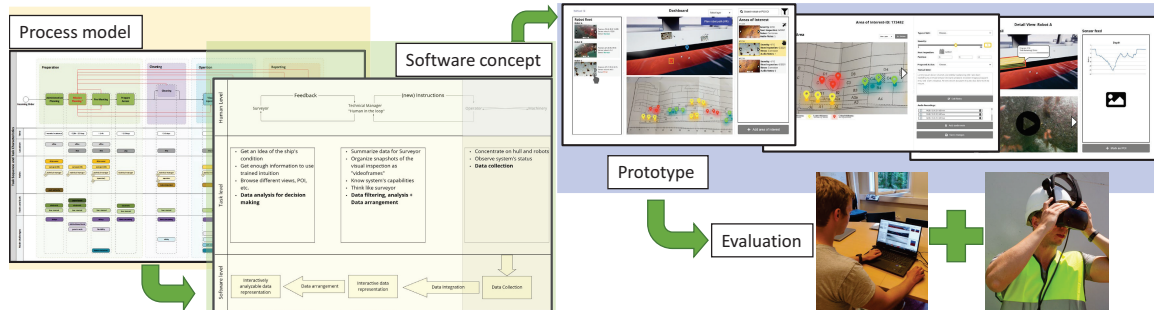


Figure 1: Outcomes along a design process for a hybrid interface. Own representations/photographs.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Where robots and humans interact, an interface is needed, that meets the demands of human control over robotics, taking into account both user expectations and task characteristics. This is especially true when interacting with an entire fleet of robots and monitoring their task execution. In order to design an appropriate user interface (UI), it is necessary to obtain precise knowledge of the working conditions in which the technology will be used [5]. Furthermore, it is indispensable to know the steps of the work process and to identify which of these steps the user interface is involved in. LaViola et al. [5] provide a wide collection of design principles fundamentally relevant, but not limited to 3D UI design. In order to meet the conditions for a useful UI, various strands must be brought together according to these design principles. For example, LaViola et al. [5] stress that in the work context a crucial rule to the design of a UI is ergonomics, considered as the process of designing the UI such that it is easy to use physically. To comply with such fundamental principles, it is necessary to stick to a process that ensures that the needed knowledge can be gathered and transferred into the design. While there are existing design process patterns, a process specifically for multi-robot user interfaces is hard to find. Therefore, the main question driving the work at hand arises: How can a design process for a multi-robot UI for a work context be implemented?

An answer to this needs to include three main aspects, which we will investigate in context with a related use case. Initially, it is necessary to identify **task characteristics** that have to be considered in UI design, such that the central requirements of the tasks can be fulfilled at a maximum of usability. The designers need to know the

circumstances that the future UI will be used in and tailor the UI to a system, whose advances and pitfalls can only be anticipated, especially when the actual robotic system is not a final product yet but may still be under active development [5].

Moreover, it is necessary to identify and characterize the relevant user roles as well as **user's needs**, involved in the tasks this UI is used for [6]. This will help to empathize with the future user group and to tailor UI characteristics to their needs.

Lastly, it is necessary to find out, how to provide the user with a sufficient **situational understanding** of the robotic system to support trust and reliability, as they are important measures in human-robot teaming [8]. It has been shown, that the use of virtual reality (VR) can be used as a tool for robot reprogramming [7] and thereby establish this kind of awareness. However, it is still to be found out if and how a VR environment can help in situations where multi-robot systems need to be directed and monitored in operation. More specific, it is necessary to identify a way how to integrate VR capability into a UI for robot mission planning and monitoring.

This work contributes by describing how to assess knowledge about task characteristics, users needs and how to provide situational understanding along the design process for a multi-robot UI based on general design principles. Additionally, it contributes by finding case specific design guidelines and principles for the use in multi-robot UI design and how to make use of VR support.

2 METHODS

To gather knowledge about the task characteristics and work processes, two reasonable approaches can be combined: Process and knowledge modeling as well as psychological work analysis. To model processes and knowledge, workers are observed at their current tasks, identifying where the robotic system can be introduced and interview the relevant stakeholders about requirements that their work needs to meet. A cognitive work analysis [4] identifies how demanding a task is and where these demands derive from. Crucial to this analysis process is the participation of end-users as their early participation in early technology developments helps raising the acceptance upon introduction [6]. The process and knowledge modeling should be visualized in a flowchart diagram to easier integrate the model into further development processes [4]. Work

*e-mail: gruendling@uni-trier.de

analysis as a contextual analysis includes e.g., work roles, social context, task analysis. This results in fundamental knowledge about task and users, represented as task and user models. Personas as user models describe the key characteristics of stereotypical users and make it easy to empathize with the end user [2]. They make it possible to adapt the technical solutions so that they pick up the users where they are with their mental process model. Personas thus provide valuable impulses for implementation and can contribute to increasing user acceptance and satisfaction [5].

Based on process model and work analysis, a software concept with initial design ideas and mock-ups are created that contain functions tailored to the user's needs in the respective role and task. User feedback is already gathered on these mock-ups in order to further involve users in the development process. The afterwards implemented prototype is iteratively evaluated and checked against the requirements of task and users (e.g. System-Usability-Scale [1]). These iterations are repeated until a termination condition is reached, for example a certain level of usability.

3 USE CASE

The following section describes a use case where the previously described process is actively applied. Cargo ships have to be inspected in recurring cycles according to the specifications of the classification society or flag state. Inspection includes that several hundred square meters of metal structures have to be tested for corrosion, bio-fouling and other damage visually and by ultrasonic non-destructive testing (NDT) [3]. This process could benefit from a multi-robot system, as it can improve measurement resolution, because more measurements across the surface can be taken at the same time while reducing risks for workers. Therefore, we are investigating, how a UI for such a use case could be designed.

The work analysis gave insights into (1) the key roles involved in the inspection process, (2) the process and task sequence of a ship inspection, and (3) interdependencies between processes phases, roles and subsequent/previous inspections. Three main work roles (personas) appeared relevant in the inspection: Operator, Technical Manager and Surveyor. Typically, a team of at least two operators manually records five measurements per square meter along a ship's hull using an ultrasonic probe. These measurements are checked by the surveyor, who then decides on the operating permission for the ship. The Technical Manager is responsible for coordinating and planning when which part of a hull will be surveyed, as well as scheduling repairs.

The inspection includes assessing the current condition of the hull and identifying the points to be inspected by performing the measurements and interpreting the recorded readings. For each of these three process steps, an interface was designed that initially provides basic information via a desktop and allows the activation of an immersive virtual environment at any time to interact with a 3D model of the ship. The interaction possibilities range from viewing and navigating around the ship to planning robot missions. For planning robot missions, the user utilizes a dedicated on-demand VR environment to specify an area on the hull that should be measured or to set up waypoints for robots. For this step of the process, the immersive three-dimensional representation is particularly helpful, since placing waypoints in mid-air is much more comfortable and easier in VR than on a PC screen. Depending on the robots available, the measurements include visual material (photos and videos), thickness and 3D Scans. During the mission, the user monitors the robots position, state and movements. After a completed mission, the data is presented to the user in a dedicated report view with possibilities to add annotations to measurements, to enable re-planning and to meet the strong interdependencies between roles.

UI designs are iteratively evaluated and continuously refined based on stakeholder interviews and end-user feedback. The first two iterations of formative evaluations were design-based, meaning

that no working prototype existed at the time. The third evaluation was based on a first working software prototype that allowed the respondents to perform the previously identified tasks in a simulated environment. Subsequently, the respondents were asked to answer questions regarding trust, usability and required knowledge. Since this is an ongoing work, the summative evaluation is still pending.

4 RESULTS

Respondents indicated that they generally trust the system and particularly emphasized that the ability to view and interact with a 3D model in VR helped to gain an awareness of the state of the overall system. Respondents also rated the function of being able to decide in VR which parts of the ship could be inspected as helpful for the technical manager as well as the surveyor. The respondents gave an average SUS score of 75, $N = 5$. While this score should obviously not be interpreted as a robust metric due to the small sample size, it does provide a good indicator of whether the direction of UI development is desirable. Participants indicated that they would welcome a greater range of functions in VR. This was mainly related to the monitoring of the multi-robot system, as the participants indicated that they expected a better spatial understanding of the state of the overall system in VR.

5 CONCLUSION

The experiences from the design process show that the UI development for a multi-robot system should focus on providing the user with the starting conditions for the tasks to be performed in a simple way (planning), as well as the possibility of recording the overall status of the system at any time (monitoring). The results from the first evaluations of our use case motivate us to provide further VR-supported functions in the interface and thus to further refine the design steps needed for a modern hybrid interface for interaction with multi-robot systems. Particularly interesting is, how the interface shapes the user's mental model of the system and its state.

ACKNOWLEDGMENTS

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No. 871260

REFERENCES

- [1] J. Brooke. SUS: A quick and dirty usability scale. *Usability Eval. Ind.*, 189, 11 1995.
- [2] F. Y. Guo, S. Shamdasani, and B. Randall. Creating Effective Personas for Product Design: Insights from a Case Study. In P. L. P. Rau, editor, *Internationalization, Design and Global Development*, volume 6775 of *Lecture Notes in Computer Science*, pages 37–46. Springer Berlin Heidelberg, 2011.
- [3] M. E. Ibrahim. Nondestructive testing and structural health monitoring of marine composite structures. In *Marine Applications of Advanced Fibre-Reinforced Composites*, pages 147–183. Elsevier, 2016.
- [4] D. P. Jenkins, N. A. Stanton, P. M. Salmon, and G. H. Walker. *Cognitive Work Analysis: Coping with Complexity*. CRC Press, 2017.
- [5] J. J. LaViola, E. Kruijff, R. P. McMahan, D. A. Bowman, and I. Poupyrev. *3D user interfaces*. Pearson always learning. Addison-Wesley, second edition edition, 2017.
- [6] A. Paruzel, D. Bentler, K. D. Schlicher, W. Nettelstroth, and G. W. Maier. Employees First, Technology Second. *Zeitschrift für Arbeits- und Organisationspsychologie A&O*, 64(1):46–57, 2020.
- [7] T. Togias, C. Gkournelos, P. Angelakis, G. Michalos, and S. Makris. Virtual reality environment for industrial robot control and path design. *Procedia CIRP*, 100:133–138, 2021.
- [8] S. You and L. Robert. Trusting Robots in Teams: Examining the Impacts of Trusting Robots on Team Performance and Satisfaction. In T. Bui, editor, *Proceedings of the 52nd Hawaii International Conference on System Sciences*, Proceedings of the Annual Hawaii International Conference on System Sciences, 2019.